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## A methodology for modelling energy use in the agriculture sector for the Pakistan integrated energy model

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**Abstract:** This paper attempts to rationalise and facilitate the application of the MARKAL/TIMES modeling framework for modeling and planning energy use in Pakistan which has been developed as the Pakistan integrated energy model (Pak-IEM) by the Government of Pakistan. The arguments are developed by applying the paradigm to the agriculture sector. This modeling approach is rooted in the fundamentals of decision modeling involving the explicit identification of values and objectives, the decision context, and the use of a logical framework to structure the elements of the decision situation through influence diagrams. This lends itself perfectly to a linear programming modeling framework. The practical implications of this are the advantages to be gained in successive refinements of the integrated model in later years, contingent to operational, policy, or technology changes. This capability, hopefully, will identify a host of potentially feasible related projects that can be undertaken within various sectors.

**Keywords:** intelligent systems; energy modelling; influence diagrams; linear programming; MARKAL/TIMES; Pakistan.

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**Biographical notes:** Sayed M. Irfan is currently employed with 7-Eleven Corporation, Canada. His previous experience includes extensive consulting experience in the energy conservation and management projects in Pakistan. He received his Bachelor's in Civil Engineering in 1981. His graduate degree obtained in Industrial Engineering and Operations Research in 2002. He pioneered the development and delivery of the graduate program in engineering management at NED University and instituted the culture of professional training from the university's forum. He brings along a diverse perspective on practical advantages to be gained through implementation of the supply chain management paradigm. He is a founding member of Supply Chain Association of Pakistan.

## 1 Introduction

The Asian Development Bank (ADB) had carried out a technical assistance (TA) program with Pakistan's Planning Commission during 2008 to 2010 to assist the Government of Pakistan (GoP) in developing an integrated energy model (Pakistan Integrated Energy Model – Pak-IEM) that will allow a national energy planning group to assess the impacts of various strategies for meeting future energy needs in an optimal manner. The model will integrate planning factors pertaining to financial, economic, energy supply, national resource, energy use, environmental impact, technology improvement, energy efficiency, and conservation to assess the costs and benefits of policies that will shape the country for the coming decades (Pak-IEM Model Design Report, 2009).

This paper attempts to systematically develop a coherent basis for justifying how and why the MARKAL/TIMES modelling environment is relevant to the Pak-IEM. It explains how the strategic context can be developed for energy planning at the macro level and then how such strategies can be logically translated into an operational framework thereby setting the direction for data collection activities. Then, the organising aspect of the data is addressed which brings to the fore the assembly of the agriculture sector templates and the VEDA-FE environment. The key inputs for generating the typical linear programming (LP) model are thus identified. Finally, a framework for establishing the key output is briefly discussed that lends itself to an optimisation solution process to obtain a *least-cost* solution in terms of an *optimal fuel mix*. This aspect is handled using the GAMS optimisation engine and the VEDA-BE.

To properly illustrate this entire rationale, the agriculture sector has been chosen primarily for its relative simplicity. Though the Pak-IEM transcends the entire spectrum of national energy consumption, which warrants the proper integration of macro-level policies, pricing, and technology issues, and demand and supply dynamics, treating the agriculture sector in isolation still serves the purpose of the characterisation of the overall model dynamics in a lucid and an effective manner.

Agriculture is one of the largest sectors of the economy in Pakistan, contributing 21.8% of GDP, and an even larger share of employment, providing an income for 44.7% of the total labour force (Pakistan Economic Survey 2008–2009). Due to the size of the sector, energy use is significant even if production is relatively non-energy intensive.

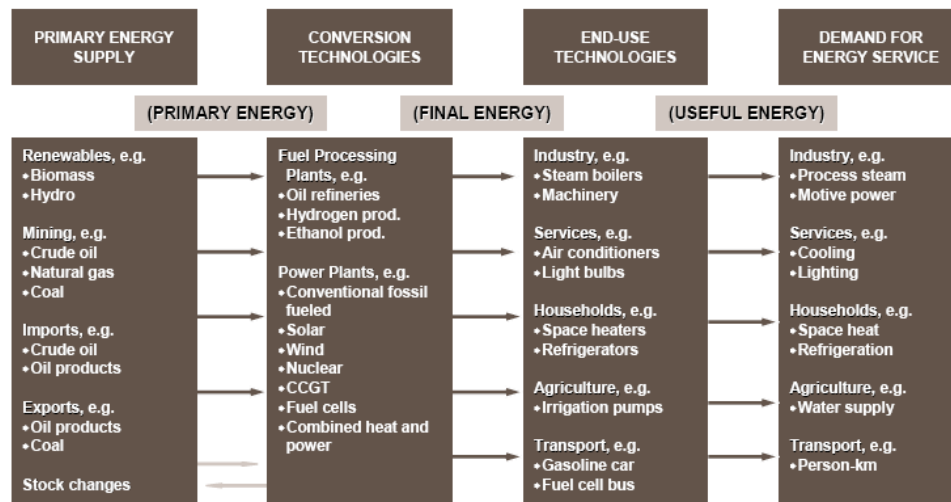
## 2 The MARKAL/TIMES modelling framework overview

A MARKAL/TIMES model requires a core set of data to characterise the energy system that the model is intended to represent. Once constructed, the model will solve for the *least-cost solution* that utilises the available energy carriers (fuels, electricity, and heat) and technologies (power plants, refineries, and end-use devices) to meet the energy service demands (e.g., irrigation water pumping, space conditioning, industrial process heat, lighting, and passenger kilometres travelled), subject to physical limitations (e.g.,

resource availability, and hydro/solar/wind potential), targets (e.g., renewables share of electricity generation) and policies (e.g., GHG limits, emission caps, nuclear), and other constraints (e.g., rate of technology adoption) imposed on the system (IRG, 2008a). A stylised representation of the underlying energy system depicted in the model is illustrated in Figure 1. ‘This representation implies the way the model perceives supply of energy to service sectoral demands’.

It is important to distinguish that MARKAL/TIMES, unlike many other models, uses the concept of energy service demand, not final energy demand. ‘This enables the model to evaluate fuel mix and device choices reflecting industry and consumer choices for meeting future service levels thus emphasising a deeper penetration into the demand environment’. There is a counter opinion, however, that such levels of penetration warrant extremely well-structured data management systems at organisational levels which may be difficult to visualise for this country. However, the MARKAL/TIMES modelling framework, by its very nature, is an evolutionary environment and with passage of time modifications will tend to evolve within well-defined and regulated environments and as such promote more disciplined data management practices.

**Figure 1** MARKAL/TIMES model overview (see online version for colours)

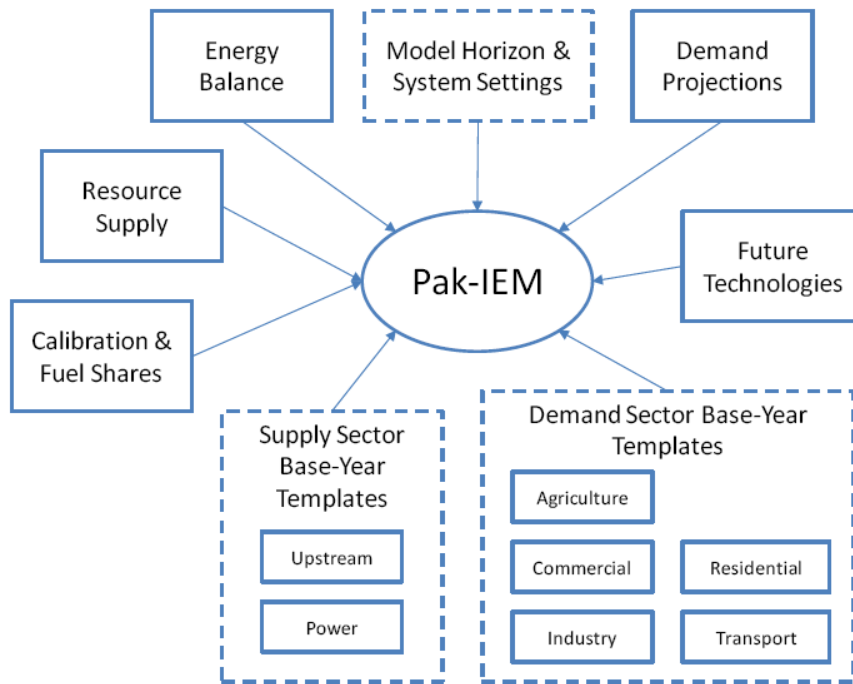


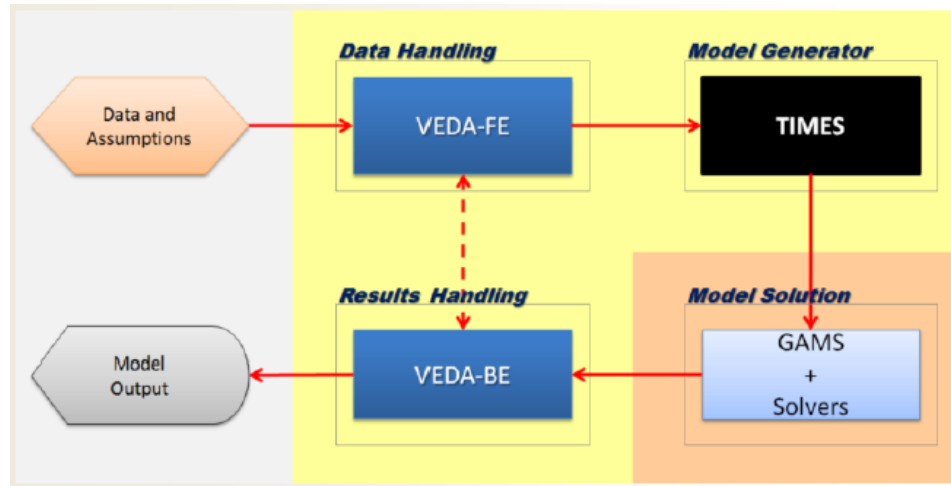
These energy service demands can be specified in whatever units are convenient (e.g., vehicle-miles travelled for transportation, lumens for lighting, crop yield per acreage, etc.). In these cases, the end-use devices must be defined in terms that convert final energy carriers into these energy service units. These energy services need to be developed as sub-sector demands, as each serves a unique (non-substitutable) demand and requires a different set of technologies (end-use devices) to provide those services. In Section 7, it will be shown how the agriculture sub-sector demands are identified and quantified.

### 3 PakIEM design overview

The Pak-IEM model is being developed using the TIMES framework, which is the successor version of the MARKAL framework supported by International Agency (IEA) and Energy Technology Systems Analysis Program (ETSAP). It will employ the VEDA model management system to organise and handle the input data and process the results (Pak-IEM Model Design Report, 2009). The development of the model input data is organised into a set of Excel workbooks (sectoral templates) that are managed by VEDA-front end (FE) to form the complete model for calibration and development of the reference scenario (see Figure 2, where boxes with dashed outlines correspond to the VEDA-FE interface). Any number of alternate scenarios can be added to the mix for future options and policy analyses. The VEDA (BE) is used for analysing the results. The VEDA/TIMES system overview is shown Figure 3 (IRG, 2009). Both the calibration and scenarios aspects with respect to the agriculture sector will be described in this paper later (Sections 7, 8, and 9, respectively).

**Figure 2** Structure of the Pak-IEM data (see online version for colours)



**Figure 3** VEDA/TIMES system (see online version for colours)

#### 4 Related work

MARKAL/TIMES is the most widely used energy systems optimisation model in the world today. It has a 30-year track record of evolution and success (IRG, 2008c). A MARKAL/TIMES application in this country does not have precedence.

Similar modelling initiatives in previous years are also, perhaps, non-existent. A recently prepared document portrays the situation as follows: In Pakistan, the first step taken to develop a reasonable data base was the 'Energy Outlook' developed by Petroleum Institute of Pakistan. However, its methodology needs considerable revision as it is based on a simple regression analysis. It does not take into account future pricing of competing fuels, and does not factor in probabilities (Economic Advisory Council, 2009).

"In our view, the Planning Commission has taken a step in the right direction by initiating an Energy Optimisation model. However, any model is as robust as the assumptions it is based on. It is therefore important that key stakeholders be involved in the development of this model and it is part of the Energy Commission's mandate" (Economic Advisory Council, 2009).

Considering the structure of the data inputs (Figure 2), it will be evident that the Pak-IEM can allow for much robustness. The calibration process with respect to a given Base Year and the comprehensive data development process will ensure that the energy supply and demand dynamics are comprehensively captured. Several documents published by various organisations (some of which are referenced in Section 11) provide a vast amount of energy and related data that can be inducted for the Pak-IEM. Quite a bit of this data can be rationalised and streamlined by engaging sector specialists and other concerned officials to refine it for the purposes of meaningful induction.

## **5 Structuring the decision context, values, and the fundamental and means objectives – *the elements of a decision model***

### *5.1 Identifying the decision context*

Each specific decision situation calls for specific objectives. The setting in which the decision occurs is the *decision context*. In the case of the Pak-IEM, the decision context can be identified as ‘the need for a long-term energy planning tool to ensure energy security at a viable cost’. This decision context will apply to all the subsequent discussions. However, certain guidelines do apply in determining whether the declared decision context is valid (Clemen and Reilly, 2001). These are:

- 1 Whether the context captures the situation at hand? Is the right problem being addressed?
- 2 Does a *decision ownership* exist?
- 3 Whether feasibility is not being compromised; that is, is sufficient time and capital available to pursue the decision?

With respect to the PakIEM, the declared decision context stands fully justified. We may observe that the current energy situation in Pakistan warrants a viable energy policy and planning initiatives. It also stands to reason that energy security being a strategic concern, the government will allocate necessary resources to develop this tool. And, finally, contractual agreements with the ADB imply that adequate time and monetary budgets have been secured to carry the project to its logical conclusion (IRG, 2008b).

When the decision context is specified and appropriate objectives aligned with the context, the decision maker knows what the situation is and exactly why he or she cares about making a decision in that situation. The next section explains how the decision context can be narrowed down from a *higher level goal* to the eventual adaptation of the MARKAL/TIMES modelling environment by a proper structuring of values and objectives, recognising that energy, as a key driver of economic progress, is the higher level goal. The argument will be intuitively as well as objectively explained by focusing on the agriculture sector, though similar reasoning will apply to the other sectors as well.

### *5.2 Structuring values*

An *objective* is a specific thing that one wants to achieve, while an individual’s objectives taken together make up his or her *values*. Values and decision context go hand in hand. The idea of a requisite model comes into play here: a requisite decision model includes all of the objectives that matter, and only those that matter, in the decision context at hand (Clemen and Reilly, 2001).

For the agriculture sector, we can start by stating that the *overall decision context* is ‘developing a long-term energy planning tool to ensure energy security in the agriculture sector at a viable cost’, which can also be thought of as the *higher-level goal*. Though this is somewhat apparent, there exists a need to follow a guided methodology, the implication being that some objectives may be relatively of a greater or lesser importance and, therefore, these need to be carefully organised. So it makes sense to consider all objectives that will impact positively on the agriculture sector of Pakistan. One way of

going about it is to generate a list of all such objectives (Keeney, 1992). Some of these may be:

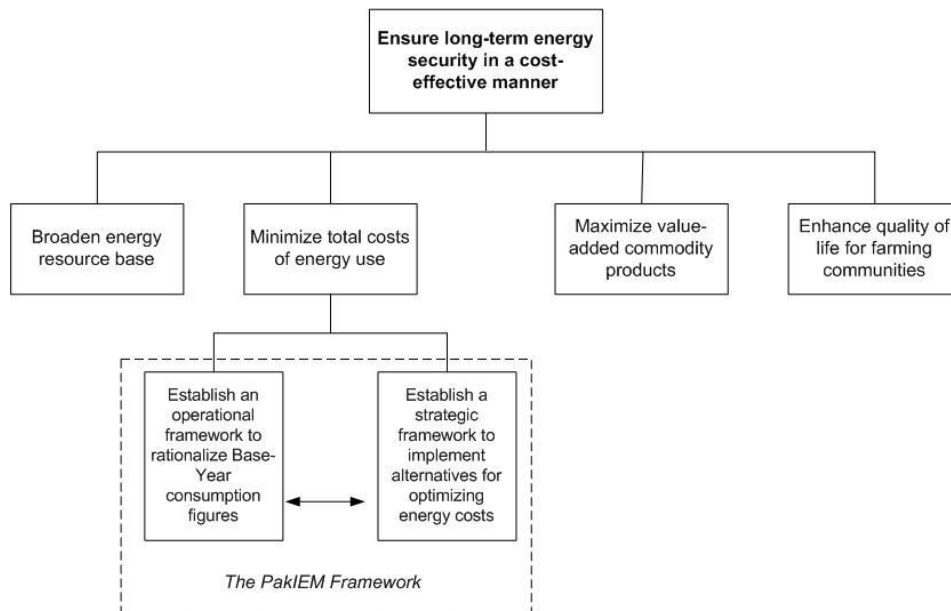
- broaden the energy resource base
- minimise total costs of energy
- maximise value-added commodity products
- achieve enhanced quality of life for agricultural communities
- provide subsidised maintenance services to farmers
- introduce more efficient farming techniques, and
- adopt more efficient technologies, etc.

This is, at best, a result of some random thinking. Though a formal way of identifying objectives is suggested by Keeney (1994), it will not be elaborated here.

### 5.3 Fundamental and means objectives

With a set of objectives that is consistent with the decision context, the next step is to separate means from fundamental objectives. This is a critical step, because here we indicate those objectives that are important because they help achieve other objectives and those that are important simply because they reflect what we really want to accomplish.

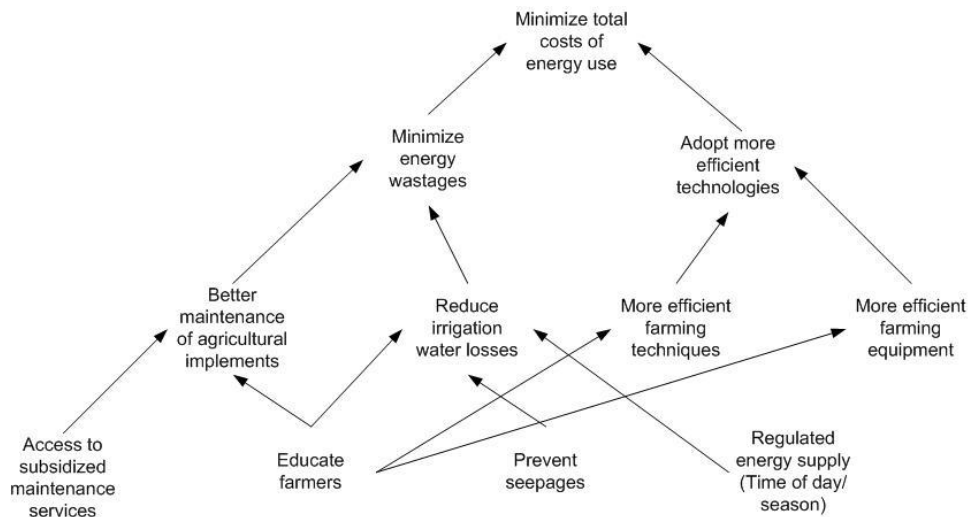
**Figure 4** The fundamental objectives



Fundamental objectives are organised into *hierarchies*. The upper levels in a hierarchy represent more general objectives, and the lower levels explain or describe important elements of the more general levels. For agriculture sector, the fundamental objectives hierarchies can be shown as in Figure 4. At the second tier are concomitant objectives that explain or describe the more important elements of the most fundamental objective. It will also be noticed that the objective ‘minimising total costs of energy use’ implies that as a first step all operating data related to energy use in the agriculture sector is identified, which will then provide the basis for analysing future requirements in a cost-effective manner.

Means objectives, on the other hand, are organised into *networks*. For the agriculture sector, once again, some of the possible means objectives can be depicted in Figure 5.

**Figure 5** The means objectives network



A key difference between this network and the fundamental objectives hierarchy is that means objectives can be connected to several objectives, indicating that they help accomplish these objectives. ‘These objectives also provide an important basis for generating creative new alternatives’. ‘This tool will invariably be very useful in the development of reference scenarios for the PakIEM’.

There is a question, however, that needs to be confronted at this stage and that is ‘How do we first separate means and fundamental objectives and then construct the two diagrams?’ A number of guiding questions are used to accomplish this task. The first question to ask regarding any objective is, ‘Why is that important?’ Known as the WITI test, this question does two things: distinguishes between means and fundamental objectives and reveals connections among the objectives. If the answer to the question is, ‘This objective is important because it helps accomplish X’, then you know that the original objective is a means objective and that it has an impact on X. Moreover, a decision maker can continue by asking, ‘Why is X important?’ By continuing to ask why the next objective is important, we can trace out the connections from one means

objective to the next until we arrive at an objective for which the answer is, ‘This objective is important just because it is important. It is one of the fundamental reasons why I care about this decision’. In this case we have identified a fundamental objective (Keeney, 1992).

Table 1 summarises four techniques for organising means and fundamental objectives. It is important to realise that one might ask these questions in any order, mixing up the sequence, jumping from the means network to the fundamental-objectives hierarchy and back again.

**Table 1** How to construct a mean-objectives networks and fundamental-objectives hierarchy

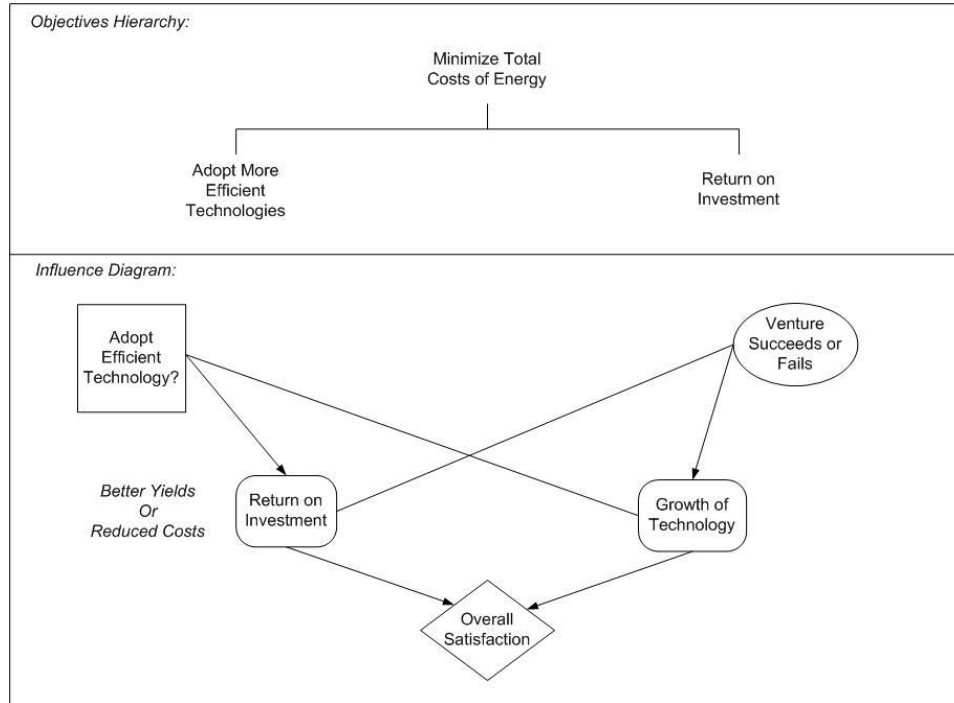
| <i>Fundamental objectives</i> |                                                     | <i>Means objectives</i>               |
|-------------------------------|-----------------------------------------------------|---------------------------------------|
| To move:                      | Downward in the hierarchy:                          | Away from the fundamental objectives: |
| Ask:                          | ‘What do you mean by that?’                         | ‘How could you achieve this?’         |
| To move:                      | Upward in the hierarchy:                            | Towards fundamental objectives:       |
| Ask:                          | ‘Of what more general objective is this an aspect?’ | ‘Why is that important?’              |

## 6 Structuring decisions

With the fundamental objectives specified, structured, and sorted out from the means objectives, the process of structuring the various elements can begin, i.e., decisions and alternatives, uncertain events and outcomes, and consequences. We can begin with influence diagrams, which provide simple graphical representations of decision situations. Different decision elements show up in the influence diagram as different shapes: rectangles represent decisions; oval represent chance events; and diamonds represent the final consequence or payoff node. A rectangle with rounded corners is used to represent a mathematical calculation or a constant value. These shapes are then linked with arrows in specific ways to show the relationships among the elements (Howard and Matheson, 1984). A relevant example could be shown as in Figure 6, which suggests that if, in the agriculture sector, more efficient technologies are adopted then, depending on market conditions or general economic conditions, ‘return on investments’ and ‘growth of technology’ *may* or *may not* feature positively to maximise overall satisfaction.

Since the PakIEM focuses on ‘developing a long-term energy planning tool to ensure energy security in the agriculture sector and at a viable cost’, the fundamental objective *minimise total costs of energy* will now be identified as the main decision context. Referring to Figure 5, we now see that all the means objectives qualify as alternatives to the objective of minimising total energy costs. The next step will be to identify appropriate data needs and evaluate such alternatives.

This process is at the heart of the MARKAL/TIMES modelling framework (Figures 1 and 2) which will now be discussed in some detail.

**Figure 6** Influence diagrams: an illustration (see online version for colours)

## 7 Modelling the agriculture sector energy consumption

The crucial first step in developing a credible integrated energy planning capability for Pakistan is acquiring and assembling the detailed data needed for a comprehensive national model. The generic data development process can be more precisely illustrated as shown in Figure 7 (Pak-IEM Model Design Report, 2009). For the purposes of this paper, however, the data development process for only the agriculture sector will be illustrated. Also, we will restrict to only a part of this development process within the elliptical region defined by the dashed lines so that only the relevant elements of the MARKAL/TIMES framework as depicted in Section 2 will be involved.

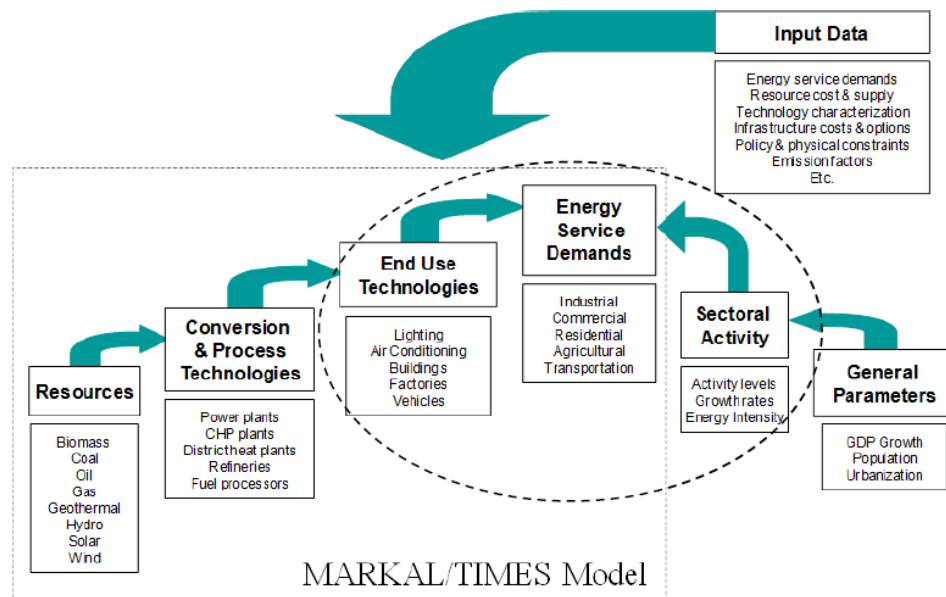
The modelling process here calls for a few things specifically; in the right sequence these are:

- 1 Identify the demands to be serviced in this sector in terms of operational activities or end-use technologies (sub-sector demands) for the base year (2006–2007).
- 2 Source the aggregate energy consumption in this sector in terms of fuel types from (Energy Year Book).
- 3 Reconcile the supply to this sector with the demands serviced with respect to specific end-use employing an appropriate approach and analysis. This step is referred to as *model calibration* and this will be done for each of the supply and demand sectors as

identified in Figure 2. This ensures that the model, when given the data, provides results that are consistent with the base-year consumption figures.

- 4 Finally, identify any number of alternate scenarios that can be added to the energy mix for future options and policy analyses. The ‘business-as-usual’ scenario can be run on the basis of projected activity levels within each sector disregarding any major policy or technology changes. On the other hand, other ‘reference scenarios’ can be analysed by bringing in policy options. As mentioned earlier in paper, this analysis carries the potential for identifying a variety of interesting and stimulating projects within various sectors.

**Figure 7** The data development process (see online version for colours)

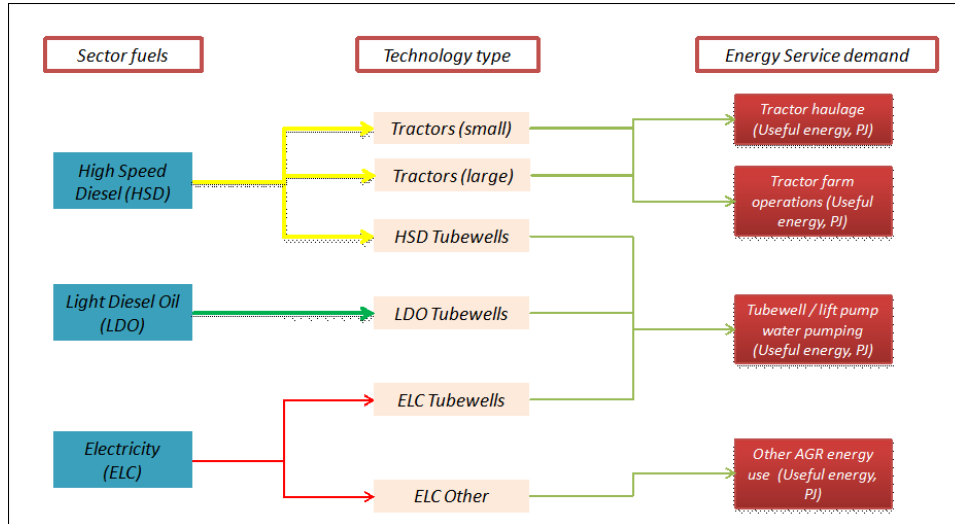


*Step 1* in this process needs to be seen in a manner consistent with the MARKAL/TIMES modelling framework. Specifically, this context is defined as the *reference energy system* (RES) (Pak-IEM Model Design Report, 2009). For the agriculture sector, this is shown in Figure 8. The energy service demands correspond to the sub-sector demands (refer Section 2) (Ministry of Petroleum and Natural Resources, 2007).

As may be seen, the main demands for energy in this sector include:

- water pumping for irrigation
- motive power provided by tractors for tilling, cultivating and harvesting
- tractor-based transport of goods
- A general ‘other’ energy use category, primarily to capture unallocated electricity use (e.g., lighting, heating and cooling for storage buildings, etc.).

(‘The RES is effectively an operational framework and it corresponds to the lowest level in the fundamental objectives hierarchy previously defined and depicted in Figure 4’.)

**Figure 8** RES for agriculture sector (see online version for colours)

*Step 2* is referenced directly to the Pakistan Energy Year Book (Ministry of Petroleum and Natural Resources, 2007). The sectoral energy use in the base year (2006–2007) totals 34 PJ, excluding any high speed diesel (HSD) use, which is included in the Energy Yearbook transport sector estimates. As shown in Figure 9, bottom-up estimates for Pak-IEM, based on technology stock data and operational data (Pakistan Farm Machinery Census, 2004; ENERCON, 1988; Van Steenberg, 1997), put agricultural HSD use at approximately 110 PJ, significantly increasing the energy consumption attributed to the agriculture sector (from 2% to 9% of final energy). Table 2 gives the energy use numbers for the base year.

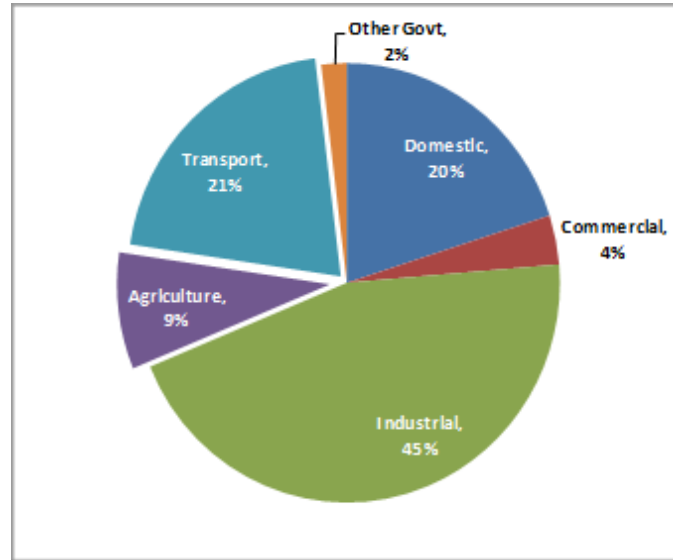
**Table 2** Energy consumption in the agriculture sector

| Source      | 2006–2007 |
|-------------|-----------|
| Oil         | 101,406   |
| Electricity | 665,861   |
| Total       | 767,266   |

In *Step 3*, the model calibration process, will now be illustrated with the help of influence diagrams. ‘Calibration’ means ensuring that the model solution replicates the 2006–2007 energy system characterisation in terms of energy production/consumption, as shown in energy balance, stock of technologies, and emissions limits.

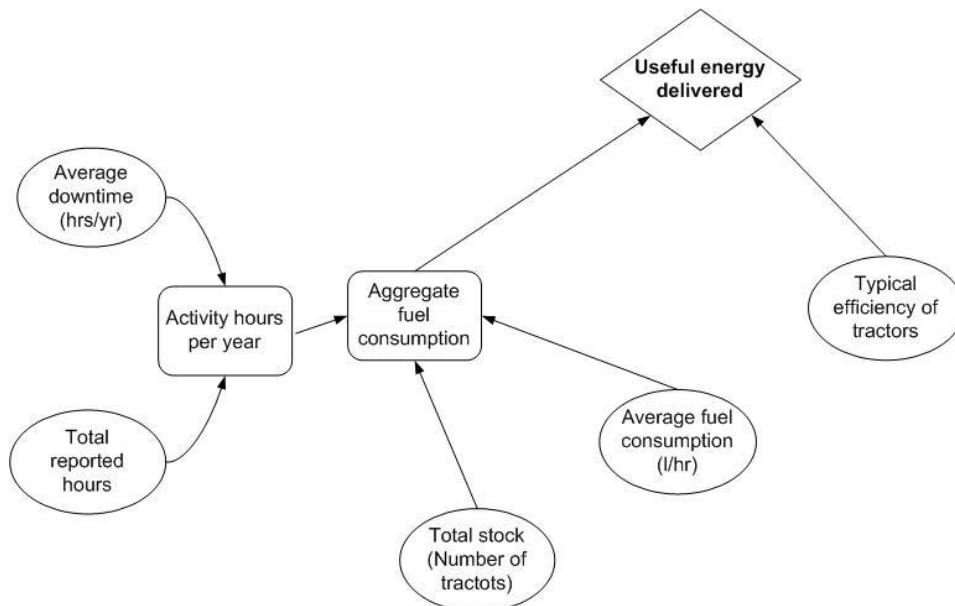
The logic of influence diagrams was introduced in Section 6. For the process in question, the *useful energy delivered by each technology* type needs to be computed; therefore, all data elements that will influence this content should typically be carefully sifted from the data sources available. This type of structuring will also be very useful when the focus shifts towards generating the ‘what-if?’ scenarios within a defined strategic framework. The influence diagrams are, in fact, a very effective counterpart when the modelling process is being implemented on spreadsheets where the data elements will correspond to specific cells. This will be demonstrated later in this section.

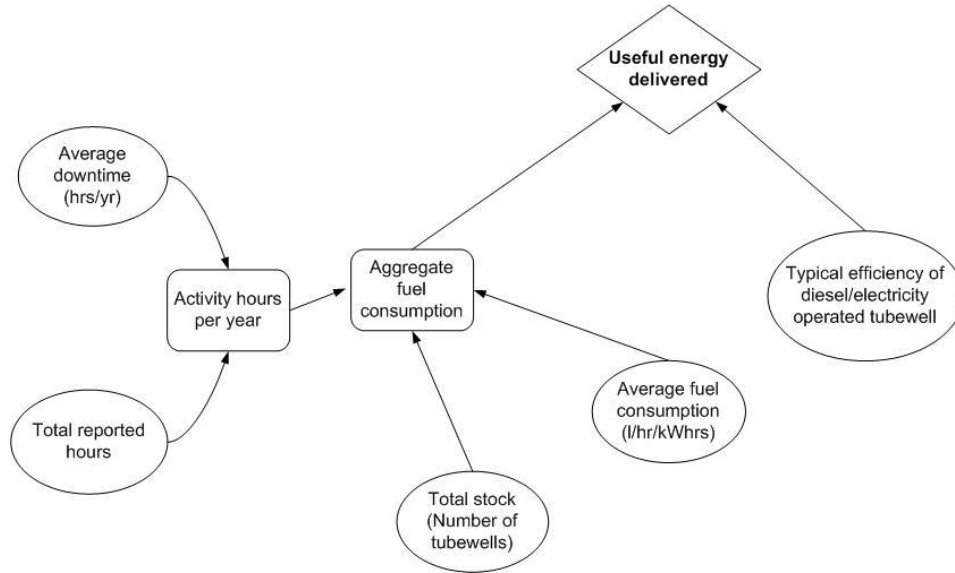
**Figure 9** Sectoral distribution of total energy consumption (see online version for colours)



Two examples will be shown here, one each for tractors and tubewells, i.e., Figures 10 and 11 (disregarding the technology types for the moment). One has to proceed in a ‘top-down’ manner by first identifying what the variable of interest is, and then decomposing each of these variables, by asking ‘What variables do I need to calculate this?’ The shapes that do not have any arcs entering into it are simple numeric data or constants (Plane, 1994).

**Figure 10** Useful energy delivered by a tractor



**Figure 11** Useful energy delivered by a tubewell

## 8 Agriculture sector templates and the VEDA (FE) data management system

As stated earlier, the Pak-IEM will employ the VEDA (FE) and (BE) model management system to organise and handle the input data and process the results. The *useful energy delivered*, as established with the help of the influence diagrams, resides in the template ‘Agriculture’ in the Excel Workbook VT\_PAK\_AGR\_v--, the screen images of which are shown in Figures 12(a) and (b). The useful energy delivered for all technology or end-use devices are shown here which can then be aggregated to define the respective sub-sectors. There is a lot of other input data that completes the assembly of the agriculture template, much of which exists in the agriculture sector workbook (identifiable as different *tabs* in the template).

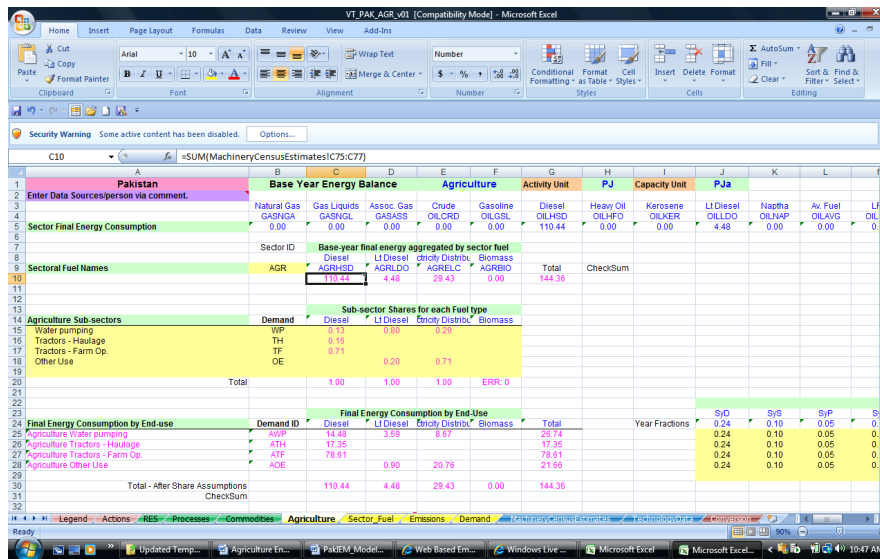
All of this is organised around the VEDA (FE) interface. The VEDA templates permit the user to organise the information in a flexible schema (Gargiulo, 2009). The key inputs are:

- characterisation of the current stock of existing technologies
- resource supply curves and cumulative resource limits
- the characterisation of future technology options
  - 1 fuels in/out, efficiency, availability, technical life duration
  - 2 investment, fixed and variable O&M costs, and ‘hurdle’ rates
  - 3 emission rates
  - 4 limits on technical potential, etc.
- demand breakdown by end-use

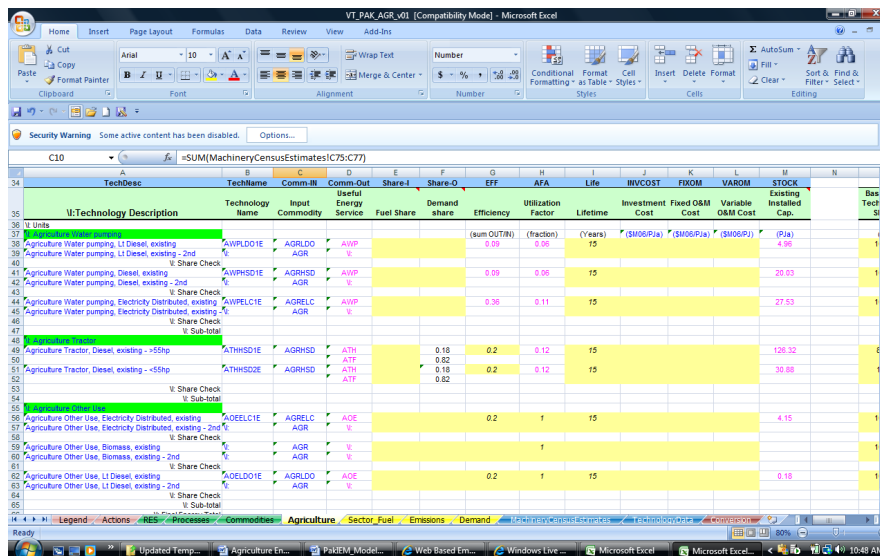
- 1 demand for useful energy
  - 2 own price (and income elasticities)
  - 3 load curve
- discount rate, reserve margin.

A screen snapshot of the VEDA (FE) input data interface is shown in Figure 13.

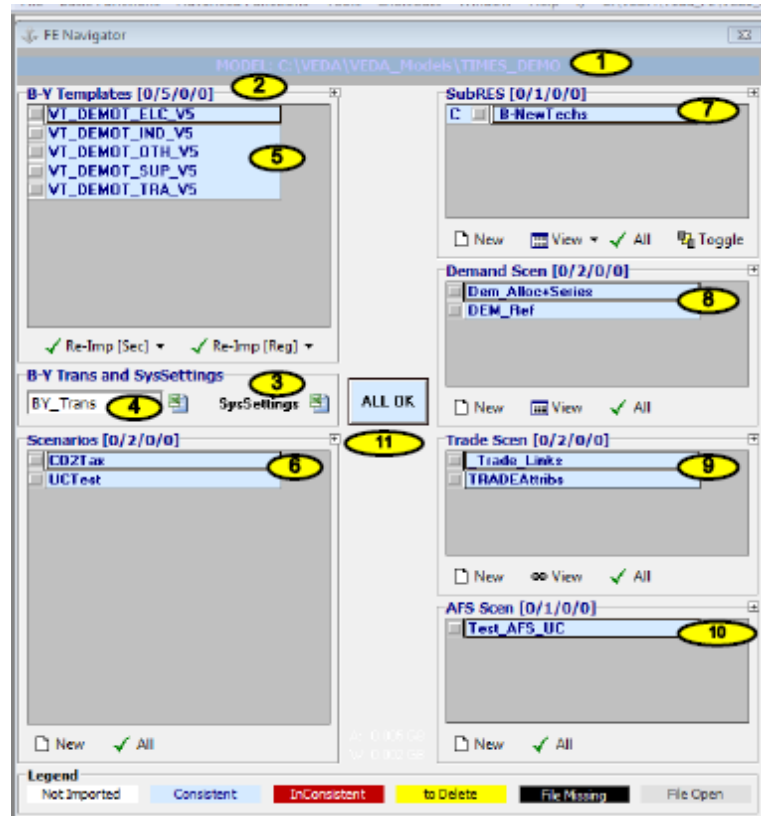
**Figure 12** (a) Agriculture sector template (b) Agriculture sector template (see online version for colours)



(a)



(b)

**Figure 13** VEDA (FE) model input data (see online version for colours)

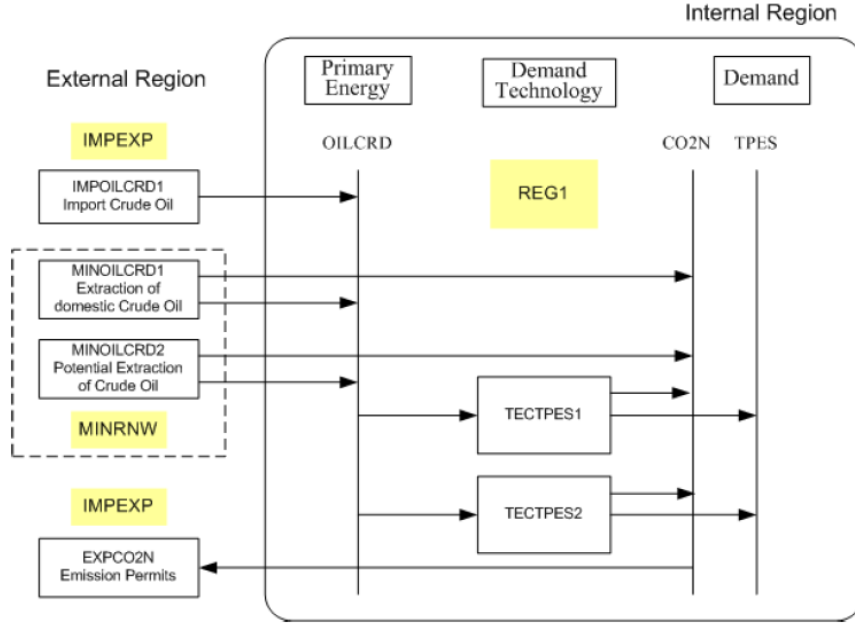
The information organised in the VEDA (FE) is in a manner that is required for the structuring of the complete Pak-IEM, a LP formulation with the objective of determining the *least-cost* energy mix to service all energy service demands (Figure 2, Section 3). This aspect of the modelling will now be discussed in Section 9 with respect to the agriculture sector subsector demands only.

## 9 Model formulation, output and the VEDA-BE data analyser

### 9.1 A general LP modelling framework

A LP modelling framework for solving the problem will now be briefly discussed. This exercise is typically known as generating and solving the TIMES model via VEDA (FE). In order to do so, we need to depict the MARKAL/TIMES modelling framework in more LP-specific terms so that the appropriate inputs from the agriculture sector templates can be related to the modelling terminology (IRG, 2009).

**Figure 14** The LP-specific RES (see online version for colours)



Though this is more representative of an overall energy system, which the Pak-IEM is geared towards, the decision elements relevant to the agriculture sector templates may be easily identified here. For example:

- Energy carriers: fuels, electricity, heat, etc.
- Technologies: power plants, refineries, end-of-use devices.
- Energy service demands: irrigations water pumping, tractor haulage, passenger kilometres, etc.
- Targets: renewables share of electricity, biomass contribution, etc.
- Policies: GHG limits, emission caps, nuclear generation, etc.
- Other constraints: rate of technology adoption (change), etc.

The general LP formulation that captures the overall sector characteristics can be expressed as follows:

$$\text{Minimise } C * X \quad (1)$$

Subject to:

$$A * X \geq b \quad (2)$$

$$E * X \geq demsev \quad (3)$$

- 1 is the total NPV of system cost
- 2 is a larger set of technical and policy constraints
- 3 is a set of service demand satisfaction constraints
- $X$  supply and demand-side decision variables (investment, capacity, activity).

Recalling that the useful energy delivered as computed in Section 7 is now the sub-sector demand to be satisfied; the model will determine the combinations of technology and fuel types that will result in the minimum-cost options. At the same time, the attainment of this *minimum* cost will be restricted by the availability of fuel types, and policy and technical constraints.

Extending the same reasoning towards the energy system upstream, the larger model can be expressed as:

$$\text{Minimise } C * X \quad (4)$$

$$A * X \geq b \quad (5)$$

$$E * X \geq findem \quad (6)$$

- 1 is the total NPV of system cost
- 2 is a large set of technical and policy constraints
- 3 is a set of final energy demand satisfaction constraints
- $X$  supply-side decision variables (investment, capacity, activity).

## 9.2 The VEDA (BE) analyser

VEDA (BE) is used for the analysis of model results. Although, this will initially be done to calibrate the model, the VEDA (BE) will primarily be exploited for developing the reference scenarios. The key outputs from such analysis are:

- total discount system cost
- resources levels and marginal costs, if constrained
- technology
  - 1 level of total installed capacity
  - 2 annual investments in new capacity and expenditure
  - 3 annual fixed and variable operating and fuel costs
  - 4 marginal cost, if constrained
- energy consumed by each technology (sub-sector), and marginal price (by season/time-of-day for electricity)
- demand marginal costs and change in levels
- emission level by resource/sector/technology for each period, and marginal costs, if limited.

## 10 Conclusions

It goes without saying that the key to economic progress is energy self-sufficiency and given the current indicators the energy crisis will only worsen in Pakistan if energy planning is not done in a systematic manner. The PakIEM is a timely initiative, though somewhat ambitious in terms of data collection efforts but this is something that should be perceived as only transient in nature. The advantages to be gained from the PakIEM will be far-reaching by virtue of a planning base that is integrated, realistic and robust.

This paper has shown how the MARKAL/TIMES modelling framework fits into the overall scheme of things and that it is consistent with the norms of any decision-making paradigm. The development and analysis of reference scenarios, in particular, can be very effectively implemented with the help of the decision modelling framework presented herein. Moreover, the recommendations made by the Energy Expert Group in the *Integrated Energy Plan 2009–2022* can be comprehensively evaluated.

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